

# Software Requirements Specification

| Field                | This example                                                                                                                                                                                                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Team name            | Dek70                                                                                                                                                                                                                     |
| Project title        | NUDTEE                                                                                                                                                                                                                    |
| SDLC chosen + reason | Agile (Scrum, feature-based sprints) — a 5-person team can divide work by functional requirement (FR-1 to FR-5), build and integrate in short iterative sprints, and adjust scope quickly given the limited build window. |
| Date / version       | v1.0 · 2026-09-25                                                                                                                                                                                                         |
| GitHub repo          | <a href="https://github.com/6931503074-sudo/NUDTEE">https://github.com/6931503074-sudo/NUDTEE</a>                                                                                                                         |

## 1. Introduction

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### 1.1 Purpose and scope

Board game players who want to find more people to play with especially people who enjoy the same genre currently rely on manual back-and-forth in LINE or Facebook groups. Existing groups do not show what genres each member enjoys, so organizers invite the wrong people or people who are not interested; nor is there a clear, live count of how many players a session still needs, so organizers must ask around one by one. This wastes time and often means a session never comes together even though enough interested players existed. NUDTEE is a small web app that lets a group show, in real time, which sessions are open, what genre each session is for, and exactly how many more players are needed so a session stops depending on someone manually counting replies in a chat thread. Success looks like this: an organizer opens the app, sees which sessions still need players and of what genre, and a member can join with one tap instead of a chat exchange.

## 1.2 Users and stakeholders

| Name / role                                           | Kind                  | What they need                                                                                              | What they fear / forbid                                                                                |
|-------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Player (primary user)                                 | primary user          | Find open sessions that match the genres they enjoy, and see live how many players are still needed         | Being invited to sessions of a genre they dislike; joining a session that turns out to already be full |
| Session organizer                                     | primary user          | Fill a session with the right number of players who like the right genre, without asking one by one in chat | Ending up short of players because interested people never saw the post in time                        |
| Board game café / shop (indirect)                     | secondary stakeholder | More regular foot traffic from groups that successfully coordinate a visit                                  | Being listed as a location without being able to control or confirm capacity                           |
| Hosting / backend provider (Supabase, Google, Vercel) | supplier              | Usage that stays within free-tier limits during the course project                                          | Sudden traffic spikes or misuse that break free-tier quotas                                            |

## 2. Overall description

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### 2.1 Product context

| Priority                            | Items                                                                                                      |
|-------------------------------------|------------------------------------------------------------------------------------------------------------|
| Must (MVP core)                     | FR-1 Create session · FR-2 Set favorite genres · FR-3 Join / leave session · FR-5 Real-time session status |
| Should                              | FR-4 Filter sessions by genre                                                                              |
| Could (next term)                   | Waitlist when a session is full · Reminder / notification system                                           |
| Won't (this term, explicit promise) | Map / GPS view , Player rating or review , Café/shop booking integration , In-app chat                     |

### 2.2 Assumptions and constraints

- Assumption: every player has a smartphone or laptop and is willing to sign in with a Google account or phone number.
- Constraint: backend limited to Supabase free tier (500 MB database, 50,000 monthly active users, projects auto-pause after 7 days of inactivity).
- Constraint: hosting on Vercel/Netlify free tier; authentication via Google OAuth (no cost).
- Constraint: Thai and English UI text; no paid API or service anywhere in the MVP.
- Constraint: personal data stored is limited to display name, preferred genres, and session participation no chat content, no payment data.
- Constraint: 5-person team, shared build window with other coursework.

### 3. Functional requirements

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#### FR-1 — Create a session (room)

**User story:**

As a player, I want to create a session with a genre, location, date/time, and desired number of players, so that others can see it and join.

**Pain this traces to:**

Sessions are currently announced informally in chat, with no structured, shared place to see what is being organized.

- Given a genre, location, date/time, and min/max player count entered, when the organizer saves it, then the session appears in the session list with those details.
- Given an existing session, when the organizer edits the details, then the change is shown immediately without creating a duplicate session.
- Error: Given a required field is empty, when the organizer saves, then nothing is saved and a message appears at that field (e.g. "Please enter a location").
- Error: Given max players is less than min players, or max is below 2, when saving, then it is rejected with a message.
- Error: Given a date/time in the past, when saving, then it is rejected with a message.

#### FR-2 — Set favorite game genres (profile tag)

**User story:**

As a player, I want to tag the genres I enjoy on my profile, so that I can be matched with sessions I actually want to join.

**Pain this traces to:**

Members of existing chat groups do not know what genre each other enjoys, so organizers invite the wrong people or people who are not interested.

- Given a list of genres, when the player selects one or more, then those genres are saved to their profile and shown when relevant.
- Given a player removes a genre tag, when they save the change, then that genre no longer appears on their profile.
- Error: Given saving fails (e.g. connection lost), then the previous tags are kept and an error message with a retry option is shown.
- Edge: Given zero genres selected, when saving, then the profile saves with no tags and the session list shows all genres.

### FR-3 — Join / leave a session

#### User story:

As a player, I want to join or leave an open session with one tap, so that the organizer does not need to count replies manually in chat.

#### Pain this traces to:

Organizers currently have no reliable running count of confirmed players and must ask one by one in a chat thread.

- Given an open session with available spots, when a player taps join, then they are added to the participant list and the remaining-spots count decreases by one.
- Given a session that is already full (current participants = max), when a player attempts to join, then the join is rejected and a full-session message is shown.
- Error: Given the session is full (current = max), when a player taps join, then the join is rejected and a "session is full" message is shown.
- Error: Given the player already joined, when they try to join again, then it is rejected with an "already joined" message and the count does not change.
- Error: Given two players tap join at the same moment with one spot left, then exactly one succeeds and the other sees the full-session message.

#### Decision logic in this FR:

This is the first FR with real decision logic: the system must compute remaining spots (max – current) and must reject a join attempt once the session is full, including handling two players tapping join at the same time when only one spot remains (only one join may succeed).

## FR-4 — Filter sessions by genre

### User story:

As a player, I want to filter the session list by genre, so that I only see sessions that match what I enjoy playing.

### Pain this traces to:

Without a genre filter, players must read every session individually to find one matching their interest, which does not scale as more sessions are posted.

- Given at least one genre filter selected, when the player applies it, then only sessions matching that genre are shown.
- Given no filter selected, when the player views the list, then all open sessions are shown.
- Error/empty: Given a filter with no matching sessions, then an empty-state message is shown with an option to clear the filter.

## FR-5 — Real-time session status display

### User story:

As a player or organizer, I want the session list to update live as people join or leave, so that the player counts I see are always accurate.

### Pain this traces to:

Player counts discussed in chat quickly go stale as people change their mind, leading organizers to recount manually and repeatedly.

- Given another user joins or leaves a session, when the change is saved, then every other user viewing the session list sees the updated count within 2 seconds, without refreshing the page.
- Given a session becomes full through this real-time update, when other users view the list, then that session is visibly marked as full.
- Error: Given a viewer loses connection, then the list shows a "reconnecting" indicator, and after reconnecting it re-syncs to the latest counts.

### Decision logic in this FR:

This FR requires synchronizing state across multiple concurrently connected clients (via a real-time subscription channel) rather than simple add/remove of a single record — the golden-thread FR carried through below.

### INVEST check

| FR   | Independent | Negotiable | Valuable | Estimable | Small   | Testable | Note                                                 |
|------|-------------|------------|----------|-----------|---------|----------|------------------------------------------------------|
| FR-1 | Yes         | Yes        | Yes      | Yes       | Yes     | Yes      |                                                      |
| FR-2 | Yes         | Yes        | Yes      | Yes       | Yes     | Yes      |                                                      |
| FR-3 | Partial     | Yes        | Yes      | Yes       | Yes     | Yes      | Needs sessions from FR-1; test with seeded sessions  |
| FR-4 | Partial     | Yes        | Yes      | Yes       | Yes     | Yes      | Needs FR-1 and FR-2 data; cut first if short on time |
| FR-5 | Partial     | Yes        | Yes      | Yes       | Partial | Yes      | Depends on FR-3 events                               |

#### 4. Non-functional requirements

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| ID    | Kind                                | Statement (testable)                                                                                                        | How we will check                                                                             |
|-------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| NFR-1 | performance                         | The session list shall reflect a join/leave change on all connected clients within 2 seconds.                               | Two devices open the same session list; one joins, time until the other updates               |
| NFR-2 | usability                           | The app shall be usable on a mobile phone screen without horizontal scrolling on any core screen.                           | Manual check on a real phone for each of the 5 core screens                                   |
| NFR-3 | capacity                            | The system shall support at least 20 concurrently connected users viewing the real-time session list without errors.        | Load test with 20 simulated concurrent connections                                            |
| NFR-4 | reliability                         | A session's remaining-spot count shall never go negative or exceed the stated maximum, even under concurrent join attempts. | Two clients tap join simultaneously on a session with one spot left; verify only one succeeds |
| NFR-5 | security /<br>privacy<br>(required) | A user shall be able to edit or delete only sessions they created; attempting to modify another user's session shall fail.  | Attempt to edit another account's session via the API directly and confirm it is rejected     |

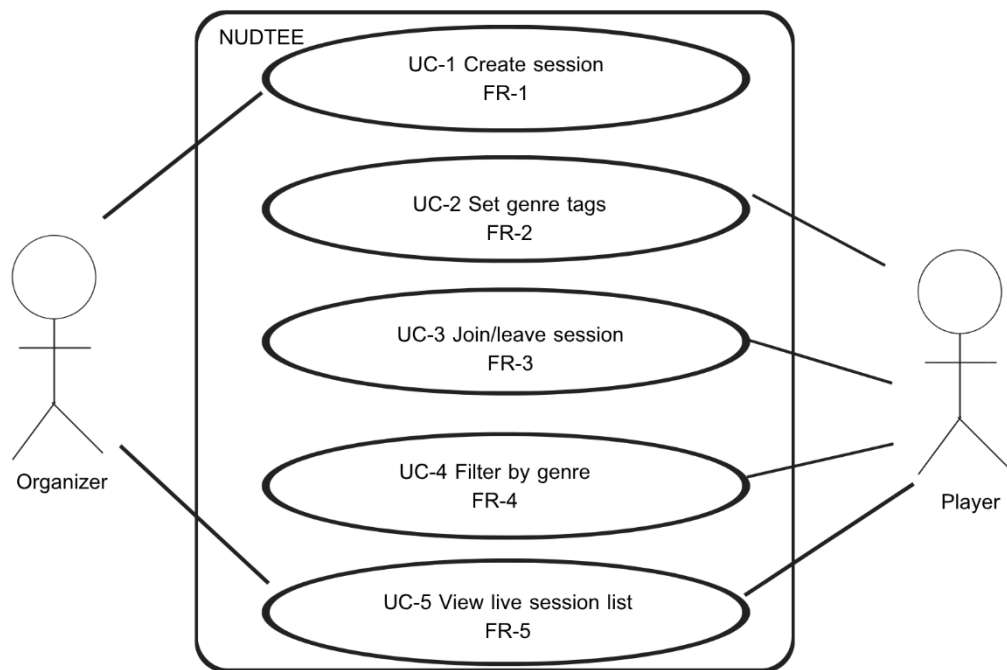
## 5. Use cases

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### 5.1 Use-case list

| Use case                    | Actor             | Related FR | Brief success story                                                       |
|-----------------------------|-------------------|------------|---------------------------------------------------------------------------|
| UC-1 Create session         | Organizer         | FR-1       | Organizer creates a "Catan, need 2 more" session; it appears in the list. |
| UC-2 Set genre tags         | Player            | FR-2       | Player tags "Strategy" and "Party" on their profile.                      |
| UC-3 Join / leave session   | Player            | FR-3       | Player joins a session; remaining-spot count drops from 2 to 1.           |
| UC-4 Filter by genre        | Player            | FR-4       | Player filters to "Strategy" and sees only matching sessions.             |
| UC-5 View live session list | Player, Organizer | FR-5       | Two players watch the same session; count updates live as a third joins.  |

## 5.2 Use-case diagram



## 6. Out of scope

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1. Map / GPS-based location view.
2. Waitlist when a session is already full.
3. Reminder or notification system.
4. Player rating or review system.
5. Integration with board game café/shop booking systems.
6. In-app chat inside a session.

## 7. Traceability matrix (Golden Thread)

| Problem / pain                                | Requirement (FR)       | Solution / design                                    | Feature built                             | Test    |
|-----------------------------------------------|------------------------|------------------------------------------------------|-------------------------------------------|---------|
| No structured place to announce sessions      | FR-1 create session    | Create-room screen                                   | Create/edit session form with validation  | (later) |
| Members' genre preferences unknown            | FR-2 genre tag         | Genre and UserGenre entities; Profile screen         | Genre tag picker on profile               | (later) |
| No reliable running headcount, asked manually | FR-3 join/leave        | join function with capacity check in one transaction | Join/leave button with remain-spots count | (later) |
| Hard to find sessions matching one's genre    | FR-4 genre filter      | Genre filter over Session                            | Genre filter chips                        | (later) |
| Player counts go stale between chat messages  | FR-5 real-time display | Supabase Realtime channel on Participation/Session   | Live-updating session list                | (later) |